

Somaiya Vidyavihar University
K J Somaiya School of Engineering

Batch: A3 Roll No.: 16010324044
Experiment / assignment / tutorial No. 05
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

TITLE: Analyse different biasing circuits of BJT

AIM: To calculate DC parameters for Fixed Bias, Collector to Base bias circuits, Voltage divider bias

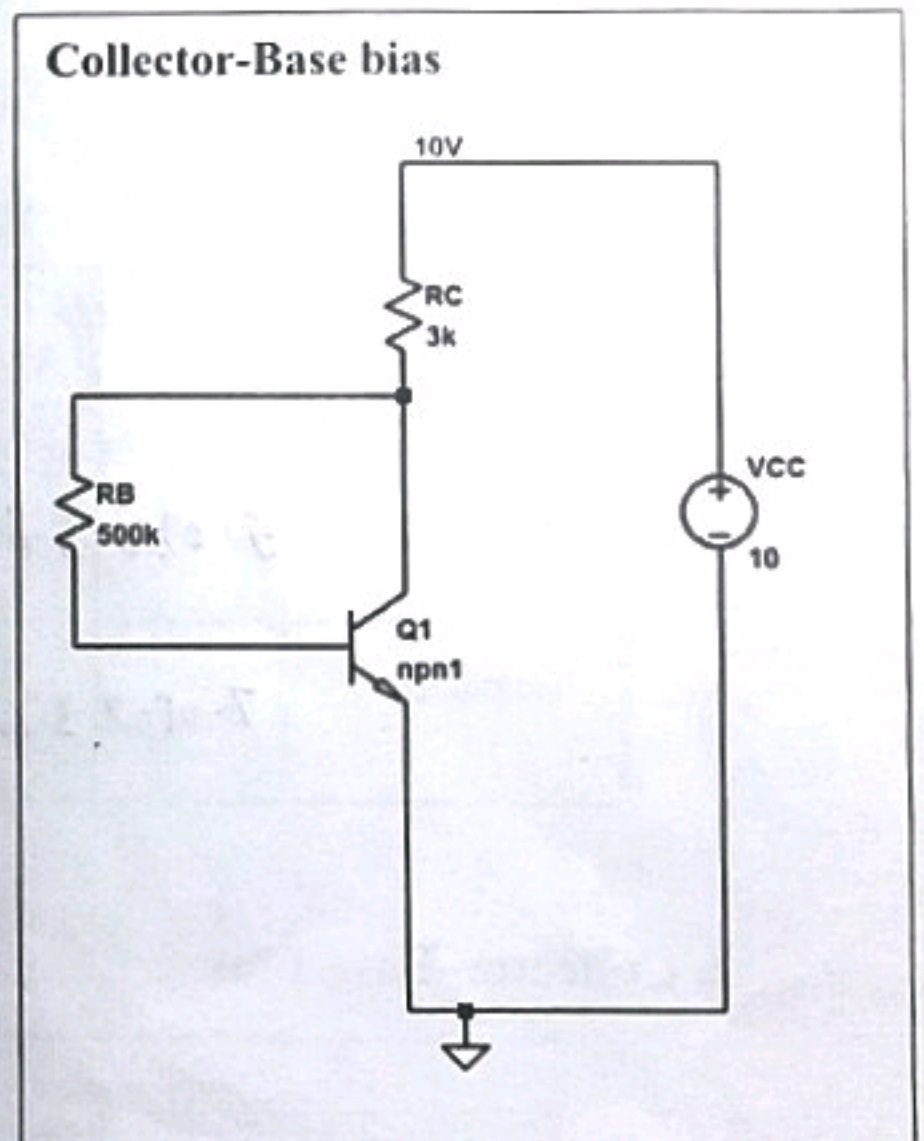
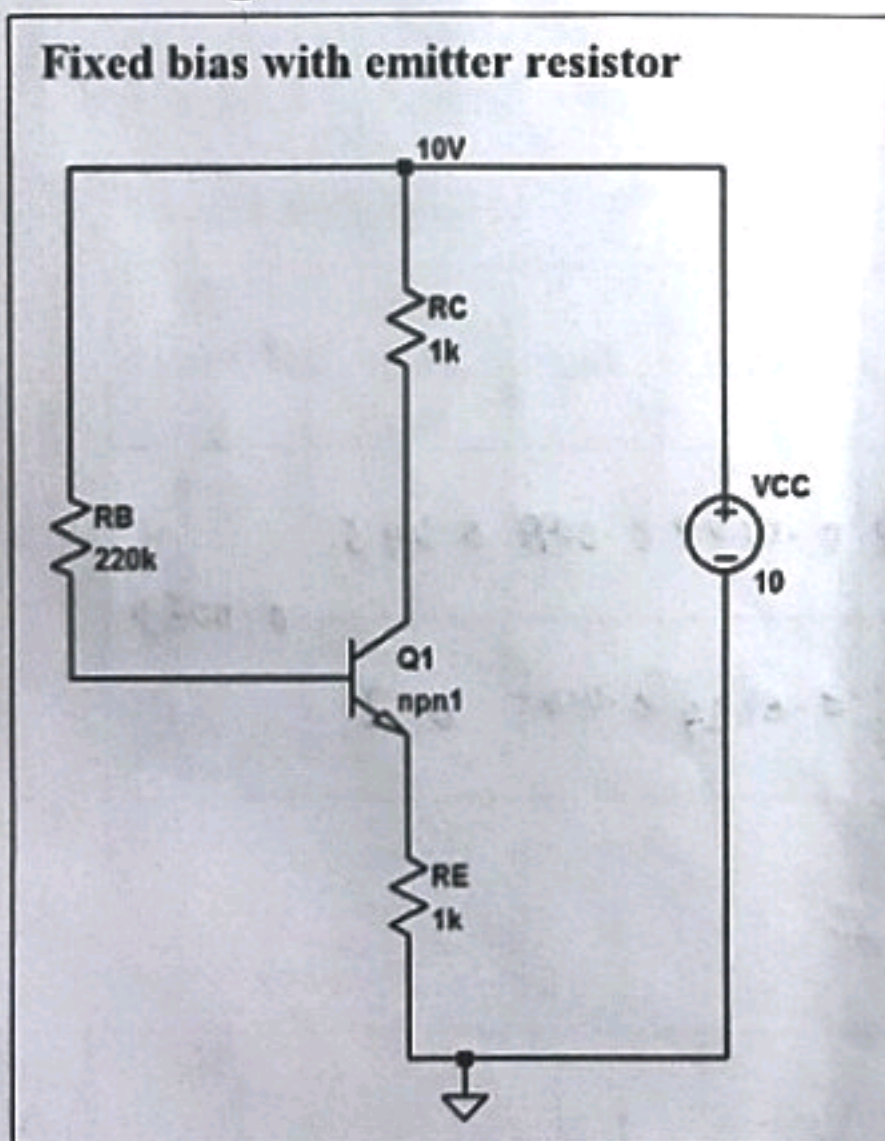
1. **OUTCOME:** Student will be able to analyze the transistor circuits for DC operation

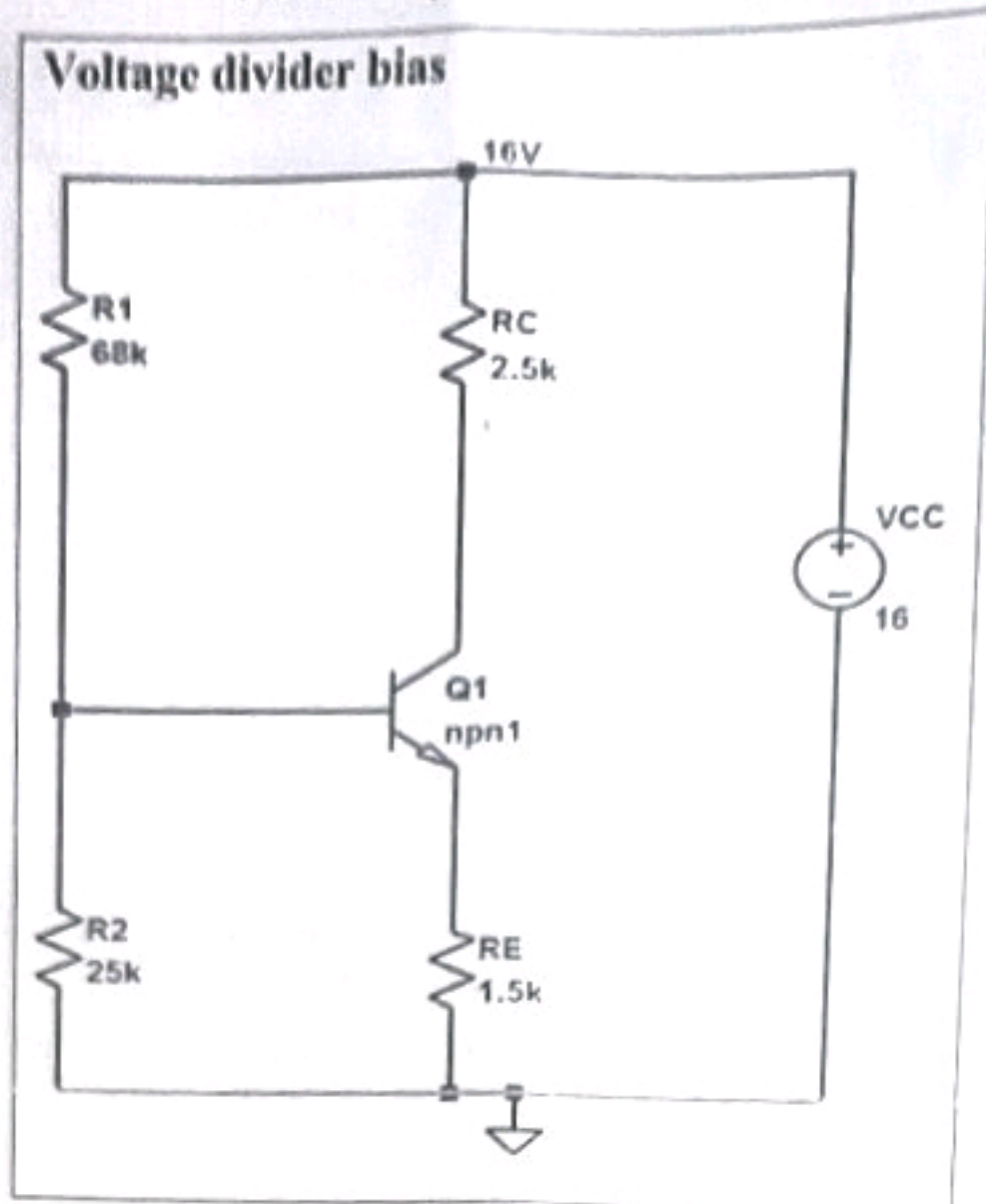
PART 1: Analysis of various biasing circuits for different values of β

Procedure:

- 1) Implement the biasing circuits in the work space of LT Spice
- 2) Perform the DC operating point analysis and find the terminal voltages and currents for different values of β (say $\beta = 100, 300$)
- 3) From the analysis, comment on the region of operation of each case. Also comment on the stability of various biasing circuits.

Circuit Diagram:





Observation Table:

1. Fixed Bias:

	V_{CEQ}	V_{BEQ}	I_{CQ}	I_{BQ}	β_{DC} (I_{CQ}/I_{BQ})	$S = \Delta I_C / \Delta \beta$
Transistor 1	5.0950	5.6082	0.0049	0.0099	0.495	0.0068
Transistor 2	7.0807	3.6156	0.0624	0.0145	0.2	

2. Collector-Base Bias

	V_{CEQ}	V_{BEQ}	I_{CQ}	I_{BQ}	β_{DC} (I_{CQ}/I_{BQ})	$S = \Delta I_C / \Delta \beta$
Transistor 1	6.463	0.6711	0.0035	0.0658	0.6034	0.0048
Transistor 2	8.3247	0.6495	0.0006	0.0077	0.2078	

3. Voltage Divider Bias:

	V_{CEQ}	V_{BEQ}	I_{CQ}	I_{BQ}	β_{DC} (I_{CQ} / I_{BQ})	$S = \Delta I_C / \Delta \beta$
Transistor 1	10.18	4.1628	0.6023	0.0378	0.0608	0.00064
Transistor 2	10.59	3.9291	0.0022	0.0102	0.2157	

Conclusion:

Voltage divider bias is the most stable among all biasing circuits, while fixed & collector-base bias are less stable and highly dependent on transistor β .

Signature of faculty in-charge